

Week 7 Worksheet Solutions

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Exercise 1. The hamiltonian for the harmonic oscillator is

$$H = \frac{1}{2m}[p^2 + (m\omega x)^2].$$

Last week, you showed that this hamiltonian can be written

$$H = \hbar\omega(a^\dagger a + 1/2),$$

where

$$a^\dagger = \frac{1}{\sqrt{2m\omega\hbar}}(m\omega x - ip)$$
$$a = \frac{1}{\sqrt{2m\omega\hbar}}(m\omega x + ip)$$

- a) Show that $[a, a^\dagger] = aa^\dagger - a^\dagger a = 1$.
- b) Show that $[a^\dagger a, a] = -a$.
- c) If $N = a^\dagger a$ and $N\psi = \lambda\psi$ for some scalar λ and some normalizable ψ , show that $Na\psi = (\lambda - 1)a\psi$.
- d) Show that if ψ is normalized, then

$$\int |a\psi|^2 dx = \int \left(\psi^* \left(\sqrt{\frac{m\omega}{2\hbar}} x \right)^2 \psi + \psi^* \left(\frac{1}{\sqrt{2m\omega\hbar}} p \right)^2 \psi \right) dx - 1/2.$$

Hint: Use integration by parts, noting that $(|\psi|^2)' = (\psi^*)'\psi + \psi^*\psi'$.

- e) Show that there is a normalized solution ψ_0 to the TISE such that $a^\dagger a\psi_0 = 0$.
Hints: Use part (d) to show that $\int |a\psi|^2 dx = \int \psi^* N\psi dx$. Together with part (c), what does this tell you about repeated applications of a to ψ ?
- f) What is the energy of ψ_0 ?

g) Show that $N = a^\dagger a$ is the **number operator**: If $\psi_n = (a^\dagger)^n \psi_0$ is the n^{th} stationary state, then

$$N\psi_n = n\psi_n.$$

h) Show that if $N\psi_0 = 0$, then a complete set of normalized solutions to the equation $N\psi = \lambda\psi$ is given by $(a^\dagger)^n \psi_0$, so that $(a^\dagger)^n \psi_0$ are also a complete set of solutions to the TISE.

Here, set

$$\alpha = \frac{m\omega}{\sqrt{2m\hbar\omega}}$$

$$\beta = \frac{1}{\sqrt{2m\hbar\omega}}.$$

a) We compute

$$[a, a^\dagger] = -2\hbar\alpha\beta = 1.$$

b) We compute

$$\begin{aligned} [N, a] &= a^\dagger a^2 - a a^\dagger a \\ &= [a^\dagger, a]a \\ &= -a. \end{aligned}$$

c) Using (b), we have

$$Na\psi = (-a + aN)\psi = (\lambda - 1)a\psi.$$

d) We can compute that

$$a\psi = \alpha x\psi - i\beta p\psi = \alpha x\psi - \hbar\beta\psi'.$$

In particular, notice that all the coefficients in front of ψ or ψ' are real. Thus, $(a\psi)^* = a\psi^*$, and we can compute

$$|a\psi|^2 = \psi^*(\alpha x)^2\psi + (\hbar\beta)^2|\psi'|^2 + \frac{x}{2}((\psi^*)'\psi + \psi^*\psi').$$

Following the hint, we can write the last term as

$$\frac{x}{2}(|\psi|^2)',$$

which can be integrated by parts to get the $-1/2$ to appear. The first term is already in the right form, while the second term can be written

$$(\hbar\beta)^2|\psi'|^2 = \beta^2[i\hbar(\psi^*)'][-i\hbar\psi'].$$

This can again be integrated by parts to pick up a minus sign and get

$$\beta^2\psi^*p^2\psi.$$

e) Part (c) means that repeated applications of a lower the eigenvalue of ψ under N . To show that

$$\int |a\psi|^2 dx = \int \psi^* N\psi dx,$$

we just need to observe that

$$a^\dagger a\psi = (\alpha x)^2\psi + (\beta p)^2\psi - \psi/2,$$

which is exactly the form of (d). Now, if $\int |a^n\psi|^2 dx = \int (a^{n-1}\psi)^* N a^{n-1}\psi dx = \int (\lambda - n - 1)|a^{n-1}\psi|^2 dx$, that means that there will be some large n such that $\lambda - n - 1 \leq 0$. Since $\int |a^n\psi|^2 dx \geq 0$, we find that $\lambda - n - 1 = 0$ for some large n . Thus, there is a state with $\lambda' = \lambda - n - 1 = 0$.

f) It's $\hbar\omega/2$, since $N\psi_0 = 0$.

g) We compute

$$N(a^\dagger)^n\psi_0 = a^\dagger a a^\dagger (a^\dagger)^{n-1}\psi_0 = (a^\dagger)^n\psi_0 + (a^\dagger)^2 a (a^\dagger)^{n-1}\psi_0 = 2(a^\dagger)^n\psi_0 + (a^\dagger)^3 a (a^\dagger)^{n-2}\psi_0.$$

At this point, we can see the pattern appearing: Eventually, we will move the a all the way to the right, which will kill ψ_0 , and this will happen precisely when the first term looks like

$$n(a^\dagger)^n\psi_0.$$

h) We already know that the "lowest" solution is ψ_0 , and we know that any $\psi_n = (a^\dagger)^n\psi_0$ is a solution. Suppose there was some other solution ψ . Since a lowers the eigenvalue of ψ by 1, there will be some large n such that $a^n\psi = 0$ and $a^{n-1}\psi \neq 0$. But then that implies $a^{n-1}\psi \propto \psi_0$. This implies that $\psi \propto \psi_{n-1}$, so we see that ψ was one of the ψ_n after all.